

The Chain

Wire Multiple Tools Into One Flow

One button. Full pipeline. Human gate preserved.

MS 3.3
Skill Sprint

Multi-Step
Orchestration

Human-in-Loop
Preserved

The Problem

Manual clicking through 4 phases. 6+ interactions. Every. Single. Morning.



Friction kills habits. 6 clicks is 5 too many for a daily routine.

The Solution

One button runs the pipeline. Stops at Approval Gate.

BEFORE:

Click Pull & Triage

Wait...

Click Draft Generation

Click Generate All Drafts

Wait...

Click Approval Gate

Review drafts

Click Send / Book



AFTER:

Click "Run Full Pipeline"

... sip coffee ...

Review drafts at Approval Gate

Click Send / Book

That's it.

Automate the mechanical. Preserve the human judgment.

What We're Building

Three additions to our existing app.py.

run_full_pipeline()

Chains fetch + triage + draft in sequence

st.status() progress UI

Live step-by-step updates as the pipeline runs

"Run Full Pipeline" button

Primary action in sidebar — one click to start

No new APIs. No new services. Just orchestration of existing code.

The Pipeline Function

run_full_pipeline() — call existing functions in sequence.

```
def run_full_pipeline():  
    # Step 1: Fetch  
    threads = load_threads(source)  
  
    # Step 2: Triage  
    grouped = triage_threads(threads)  
  
    # Step 3: Draft  
    actionable = grouped["urgent"] + grouped["needs-reply"]  
    for thread in actionable:  
        draft = draft_reply(thread)  
        session_state.drafts[thread.id] = draft
```

Same functions the buttons call — just wired together.

Pipeline Progress UI

st.status() — live updates as each step executes.

```
with st.status("Running pipeline...", expanded=True) as status:
```

```
    status.update(label="Step 1/3: Fetching threads...")
```

```
    st.write("Fetched 7 threads from Gmail")
```

```
    status.update(label="Step 2/3: Triaging...")
```

```
    st.write("Triaged: 2 urgent, 3 needs-reply")
```

```
    status.update(label="Step 3/3: Generating drafts...")
```

```
    st.write("Draft 1/4: Re: Q3 Roadmap... done")
```

```
    st.write("Draft 2/4: Re: Incident... done")
```

Users see exactly what's happening. No blank spinners. Transparency builds trust.

```
    status.update(label="Pipeline complete!", state="complete")
```

Human Still in the Loop

The pipeline automates the mechanical. YOU control the irreversible.

AUTOMATED (pipeline runs these)

- ✓ Fetch threads from Gmail
- ✓ Triage by priority (Gemini)
- ✓ Generate drafts (Gemini)

G
A
T
E

HUMAN (you decide these)

- ⚡ Approve / reject each draft
- ⚡ Click Send (irreversible)
- ⚡ Click Book Meeting (commits time)

Rule: automate the reversible, gate the irreversible.

Live Demo

One click → full pipeline → approve → send

Pipeline execution:

1. Click "Run Full Pipeline"
2. Watch: Fetching... Triaging... Drafting...
3. Auto-navigate to Approval Gate
4. Review 4 drafts
5. Approve + Send
6. Done in < 2 minutes

Pipeline log:

- ✓ Fetched 7 threads
- ✓ Triaged: 2 urgent, 2 needs-reply
- ✓ Draft 1/4: Q3 Roadmap... done
- ✓ Draft 2/4: Incident... done
- ✓ Draft 3/4: Design Review... done
- ✓ Draft 4/4: Offsite... done
- ✓ Pipeline complete! 4 drafts ready.

From inbox chaos to reviewed drafts — one button, one coffee.

What You Did Today



Created `run_full_pipeline()` — chains fetch + triage + draft



Added `st.status()` progress UI — live step-by-step updates



Added "Run Full Pipeline" button in the sidebar



Auto-navigate to Approval Gate after pipeline completes



Discussed automation vs. human-in-the-loop design

Saturday

SPRINT BUILD 3

Deploy the Full Agent

Everything from this sprint — Send + Book + Pipeline — packaged and polished.

Record a demo video of the full pipeline in action.



Screenshot your pipeline log → post on LinkedIn → [#MyAIChiefOfStaff](#)